

# LUKE SCHUMACHER

AI Research Engineer | Multimodal Data Fusion · Digital Twins · Agentic Systems

## CONTACT


Aachen, Germany
+49 151 29526325
luke.schumacher777@gmail.com
luke-schumacher.github.io/Portfolio
github.com/luke-schumacher
linkedin.com/in/luke-schumacher


## LANGUAGES

| | |
|---|---|
| **English** | Native/C2 |
| **German** | Native |
| **French** | A2 - Conversational |

## CORE SKILLS

**Programming**
Python • Java • C# • JavaScript • SQL • HTML/CSS

**ML Frameworks**
TensorFlow • PyTorch • Keras • Scikit-learn • XGBoost • Predictive Modeling

**Data Science**
Pandas • NumPy • PySpark • Telemetry & Time-Series • Signal Processing • Tableau

**Agentic AI**
Multi-Agent Systems • Neurosymbolic AI • Query Planning • RAG • LLM Orchestration • LangChain • Transformers

**Cloud & DevOps**
AWS (Certified) • Azure • Docker • Kubernetes • Git • Terraform

**Design & Database**
Figma • Adobe XD • PostgreSQL • MongoDB • Redis • MySQL

**MLOps & Industrial**
MLflow • Weights & Biases • Digital Twins • Sensor Fusion • CI/CD • Agile

## PROFESSIONAL PROFILE

AI Researcher and Technology Specialist with a strong international background spanning the Middle East, Europe, and Asia. Bachelor's in CMD & IT (Hanze, Groningen), AI & ML specialisation at Hanyang University in Seoul, and an MSc in **Artificial Intelligence for Sustainable Societies** through the prestigious Erasmus Mundus programme, studied across Tallinn, Porto, and Tampere. Now a **doctoral researcher** in Cyberphysical Systems at FH Aachen and **Project Lead for RaceIQ**, a €310K ZIM-funded R&D consortium. Five years of applied engineering — cloud infrastructure in the UAE (AWS Certified), industrial ML at infoteam, and regulated healthcare AI at Siemens Healthineers — now underpin research in multimodal data fusion, digital twins, and agentic systems. Passionate about applying AI to societal challenges and sustainable innovation, and about leveraging data to craft solutions that bridge technology and human needs. Thrives in collaborative, multicultural environments. Native German and English.

## WORK EXPERIENCE

**Research Associate | Project Lead — RaceIQ** — **Jul 2026 – Present**
*FH Aachen — Institut für Digitalisierung Aachen* — Aachen, Germany

- Leading the FH Aachen side of **RaceIQ**, a **€310K ZIM-funded R&D project (2026–2028)** building a conversational AI assistant that turns motorsport telemetry, session reports, and driver debriefs into root-cause answers with verifiable evidence.
- Own multimodal data fusion, the digital twin, and a dialogue-capable query planner that answers plain-language engineering questions from actually executed analyses rather than model-generated guesses.
- Coordinate consortium delivery with Ximaj IT-Solutions, W&S Motorsport, and Umwelt-Campus Birkenfeld in the ZIM network *Smart Green Robotics*, with Prof. Dr.-Ing. Guido Dartmann (FB05).

**Data Science & ML Engineer** — **Oct 2023 – Aug 2026**
*Siemens Healthineers* — Germany (Remote)

- Architected predictive digital-twin and anomaly detection models (Python, PySpark, SQL) optimising workflow efficiency for a global clinical clientele.
- Engineered end-to-end data pipelines integrating heterogeneous healthcare operational data into real-time decision-support signals.
- Deployed ML algorithms to production in CI/CD containerised environments (**1st Place, ITT Summit**), coordinating engineering, data science, and business stakeholders in agile sprints and presenting recommendations to senior management.

**Machine Learning Engineer** — **Jan 2022 – Oct 2023**
*Infoteam Software AG* — Germany

- Developed and deployed ML pipelines for industrial B2B clients, transforming raw production data into predictive models for process optimisation and quality control.
- Applied virtual prototyping and predictive maintenance techniques to streamline manufacturing workflows and reduce operational downtime.
- Delivered production-ready ML solutions applying clean code principles, peer code reviews, and architectural best practices in agile teams.

**Cloud Infrastructure Engineer** — **Jan 2021 – Jan 2022**
*Zero & One – AWS Partner* — UAE

- Deployed and managed scalable AWS cloud infrastructure (EC2, S3, Lambda) for diverse client projects, driving measurable operational efficiency improvements.
- Built automated CI/CD pipelines and infrastructure-as-code with Terraform across client environments; attained **AWS Cloud Practitioner Certification**.

## CERTIFICATES & AWARDS

**AWS Certified Cloud Practitioner**
*Amazon Web Services*
**ITT Summit – Siemens Healthineers**
*First Place with Georg Goertler and Belen Rodriguez*

## KEY HIGHLIGHTS

- Top 5% performance in ML & Data Science courses
- Leading global healthcare optimization project at Siemens
- 4+ years combined experience in AI/ML and cloud technologies
- International experience across 3 continents

## EDUCATION

**Doctoral Researcher – Cyberphysical Systems** **Jul 2026 – Present**
*FH Aachen / Promotionskolleg NRW* Aachen, Germany
Doctoral research on conversational, evidence-grounded AI for engineering diagnostics — multimodal fusion, digital twins, and verifiable query planning. Supervised by **Prof. Dr.-Ing. Guido Dartmann** (Cyberphysical Systems, FB05), conducted within the RaceIQ project and the ZIM innovation network *Smart Green Robotics*.

**Master of Science – AI for Sustainable Societies** **Aug 2024 – Jun 2026**
*University of Tallinn, Lusófona and Tampere* Tallinn, Porto, Tampere
Prestigious Erasmus Mundus program. **Thesis:** Autonomous Protocols and Neurosymbolic Reliability in Multi-Agent Systems for critical infrastructure, in collaboration with Deutsche Telekom/Illigo and Siemens Healthineers. Defended May 2026.

**AI & ML Specialisation – Top 5% of Class** **Feb – Sep 2022**
*Hanyang University* Seoul, South Korea
Ranked top 5% in Machine Learning and AI+X: Introduction to Data Science. Final project: ML-based cross-industry stock market fluctuation analysis. Intensive mathematical and engineering coursework underpinning the career pivot into AI.

**Bachelor – IT & CMD** **Aug 2019 – Aug 2024**
*Hanze University of Applied Sciences* Groningen, Netherlands
Technical specialisation in digital systems, HCI, and iterative software product development. International exchange programs across Europe and Asia, with self-directed pivot into AI and machine learning.

## PROFESSIONAL COMPETENCIES

**Technical Excellence:** Data-driven decision making • Statistical modeling • Algorithm design & optimization • Model deployment & monitoring • Cloud architecture design

**Collaboration:** Cross-functional teamwork • Stakeholder management • Agile methodologies • Technical communication • Knowledge sharing

**Leadership:** Project & work-package ownership • Consortium and partner coordination • Public grant delivery (ZIM) • Scrum planning • Mentoring

**Soft Skills:** Adaptability • Problem-solving • Attention to detail • Analytical thinking • Cultural sensitivity • Competitive drive

## REFERENCES

**Dr. Stefano Signoriello** – *Lead Data Scientist & Dozent (Mathematics & AI), infoteam Software AG*
Stefano.Signoriello@infoteam.de • www.linkedin.com/in/dr-stefano-signoriello

**Iolanda Vehlo** – *Assistant Professor, Universidade Lusófona*
iolandavelho@gmail.com • www.linkedin.com/in/iolanda-velho-9a594a46

**Casey Bennett** – *Associate Professor, Hanyang University & DePaul University*
cabennet@hanyang.ac.kr • www.caseybennett.com

**Georg Goertler** – *Project Manager, Siemens Healthineers*
georg.goertler@siemens-healthineers.com